
Small vs. Large Language Models: An Empirical Analysis of Accuracy, Latency, and Computational Cost

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Abstract

Large Language Models (LLMs) tend to be quite accurate in the accuracy of their information provided. However, they tend to have higher latency and computational cost as opposed to Small Language Models (SLMs), which are more inclined to be less accurate but have less latency and cost. This paper utilizes these 3 parameters of language models and evaluates them against the AMC12, an American High-School mathematics exam, and AMC8, an American Middle-School mathematics exam, on 3 separate metrics to determine how well the models fare on such tasks and when each should generally be used. On the AMC 12, SLMs achieved up to a 39.3% reduction in latency and a 49.8% reduction in token consumption; however, a steep capability threshold emerged, with the LLM drastically outperforming SLMs in accuracy 72% to 32%, a 40% accuracy differential. Conversely, on the lower-complexity AMC 8, the accuracy gap narrowed substantially from 80% and 56%, a 24% accuracy differential, while the optimal SLM achieved a 53.2% reduction in latency and a 76.9% reduction in token footprint. These findings show that while LLMs may still dominate higher level mathematical reasoning tasks due to their extreme accuracy advantage, SLMs emerge as a more efficient option for foundational mathematical tasks.

1 Introduction

Artificial Intelligence as a whole has experienced exponential adoption, with LLMs such as ChatGPT and Gemini becoming an integral part of a system for complex reasoning and problem-solving. With each passing day, these models are constantly being developed to further improve their capabilities. The common trend for improving Artificial Intelligence and Language Models is by increasing parameter sizes, resulting in a model being able to represent more variables and non-linear relationships, effectively increasing response accuracy. While it is possible to avoid issues such as overfitting or hallucinations, there is an inevitable opportunity cost that comes with an increase in parameters; steeper computational costs, high-inference latency, and significant environmental impacts such as water consumption and energy required to maintain the models' increased size to name a few. For this reason, there stems an overall argument over the effectiveness of models.

For tasks that do not require extremely complex reasoning, SLMs provide a unique alternative. SLMs generally tend to exhibit a lower accuracy than LLMs due to their parameter sizes being smaller; for instance, an average SLM contains approximately 3 billion parameters while an average LLM contains approximately 300 billion. However, SLMs offer distinct advantages, such as having a lower cost to run and being able to be run on local hardware to name a few.

Previous research and literature have defined various methods documenting possible compression techniques including quantization and pruning in order to create SLMs as well as various heuristics

where assuming simpler tasks generally suit SLMs while complex ones support LLMs. However, there remains a critical gap in quantitative frameworks defining an empirical threshold where the efficiency gains of SLMs outweigh the performance gains of LLMs.

This study addresses this gap by conducting an empirical analysis across three core evaluation metrics (accuracy, latency, and computational cost) using the AMC8 (a middle-school level competitive mathematics exam) and AMC12 (a high-school level competitive mathematics exam) as standardized benchmarks against three representative models:

- **Gemini 3.6** – A multi-purpose LLM
- **Phi-4-Mini** – An SLM designed specifically for mathematical and logical reasoning
- **Llama3.2:3B** – An SLM designed specifically for textual summaries and multi-lingual dialogue

The remainder of the paper is structured as follows:

- **Section 2:** Related literature on Language Model efficiency and capabilities
- **Section 3:** Provides experimental setup and defines evaluation metrics
- **Section 4:** Results and graphs from experiment
- **Section 5:** Discussion and limitations
- **Section 6:** Conclusion and future works

2 Literature Review

While LLMs tend to achieve high baseline accuracy through various complex tasks, SLMs have emerged as an extremely viable option in certain scenarios, optimizing latency and computational cost [1], though often at the cost of raw accuracy [1, 2]. SLMs are currently being developed through post-training compression techniques (quantization and knowledge distillation to name a few) to further optimize parameter-to-accuracy efficiency [2, 6, 7, 8]. If these SLMs are then trained on high-quality unbiased datasets, they can rival and even at times exceed baseline capabilities of larger models [4, 9, 10]: an example of this is TinyLlama, containing only 1.1B parameters, yet through optimized training and attention mechanisms, exceeds the capabilities of OPT and Pythia [5]. In addition, SLMs have the capability of being deployed on local infrastructure due to their small size, mitigating data privacy risks present through LLMs and their cloud-hosted web servers [3]. Despite these optimization breakthroughs, there still remains the issue of the empirical trade-offs between accuracy, latency, and computational cost of SLMs and LLMs across a varying set of problem/task complexities [2, 12, 13]. Finding precise accuracy-to-resource boundaries is essential for identifying specific points as to where SLMs remain optimal compared to LLMs [2, 4].

3 Methodology

This section details the specific methodology and experimental design utilized to conduct this benchmarking.

3.1 Experimental Design

This experiment was conducted with the use of 2 American Mathematics Competition (AMC) exams: AMC12B 2026 and AMC8 2026. The AMC12 is a 25 question American High School advanced Mathematics exam for top mathematics students potentially looking to qualify to the International Mathematics Olympiad (IMO), featuring complex geometry and algebraic manipulation, far more extensive than simple high-school courses. Similarly, the AMC8 is a 25 question advanced Mathematics exam for American students in Middle School, containing intermediate-level algebra and pattern recognition. For this experiment, the AMC12 is used as a hard benchmark to test the models on while the AMC8 is used as an easier one.

A total of 3 models were used to run this experiment; Gemini 3.6, Phi-4-Mini, and Llama3.2:3B. Gemini 3.6 serves as a traditional LLM, utilized for a variety of tasks such as writing text, analyzing

images, generating code, and much more. This will serve as our standard multi-purpose LLM. Phi-4-Mini is an SLM designed to excel in mathematical and logical reasoning; in short, Phi-4-Mini is specifically designed to tackle mathematics problems, and thus, the AMC8 and AMC12. Because of this niche use, Phi-4-Mini was selected as an SLM to benchmark. Finally Llama3.2:3B is an SLM designed to excel in providing textual summaries and multilingual dialogue. Because this SLM was not designed for solving Mathematics exams, it was used as a third model to show the difference between a random SLM and an SLM whose niche is designed for a specific task.

It is important to note that the prompts provided for each model should be exact copies of each other, as otherwise, differences in model outputs could be attributed towards the prompt. A few additional designs for the prompt included the temperature being set to the minimal value in order to mitigate the randomness of results provided and for the models to not use any external resources such as the internet to potentially find an answer key online instead of solving the problems with their pure intellect. With these necessities in mind, the following prompt was injected into each model:

“Prompt: You are solving a complete mathematics exam. Please follow these instructions carefully: Solve each question independently. Temperature = 0. Use only information provided in the exam. DO NOT use any external tools, calculators, internet searches, or additional resources. If you are uncertain, provide your best answer rather than leaving a question unanswered. For each question, provide one sentence explaining your reasoning, followed by your final answer. Use the exact format below for every question: Question (n) Reasoning: One concise sentence explaining your approach. Final Answer: (letter choice). Continue this format until the final question. After the final question, do not include any additional summary or explanation. Provide final token count. The exam problems are attached.”

The questions for said exams would then be attached as a text document for the models to analyze.

3.2 Experimental Settings and Evaluation Metrics

For this experiment, cloud-hosted LLM API infrastructure was utilized for Gemini 3.6, with requests executed via HTTPS API calls directly to Google Cloud endpoints. Open-weights SLMs were deployed locally via the Ollama execution engine. The computational hardware specifications for local deployment and benchmarking are detailed in the table below.

Hardware Specifications for Local Execution Environment

Component	Minimum Requirement	Recommended Specification
GPU Memory	4 GB VRAM	8 GB VRAM
System RAM	8 GB	16 GB
Storage Space	15 GB SSD	25 GB SSD

For this experiment, accuracy, latency, and computational cost were the metrics used for evaluation. Accuracy was measured in the overall percentage of questions a model scored correct on. Since both exams had 25 questions, each question was weighted as 4% out of a total 100%. A higher accuracy, *ceteris paribus*, represents a model performing better. Latency was measured as the total amount of time it takes for the models to respond, measured from the moment the prompt was injected into the model to the time it took for the model to complete its response. A lower latency, *ceteris paribus*, represents a model performing better. Computational cost was measured by the token cost of the output of the model, with a higher token cost indicating a greater cost. A lower computational cost, *ceteris paribus*, represents a model performing better. These 3 metrics were then analyzed to determine an appropriate conclusion.

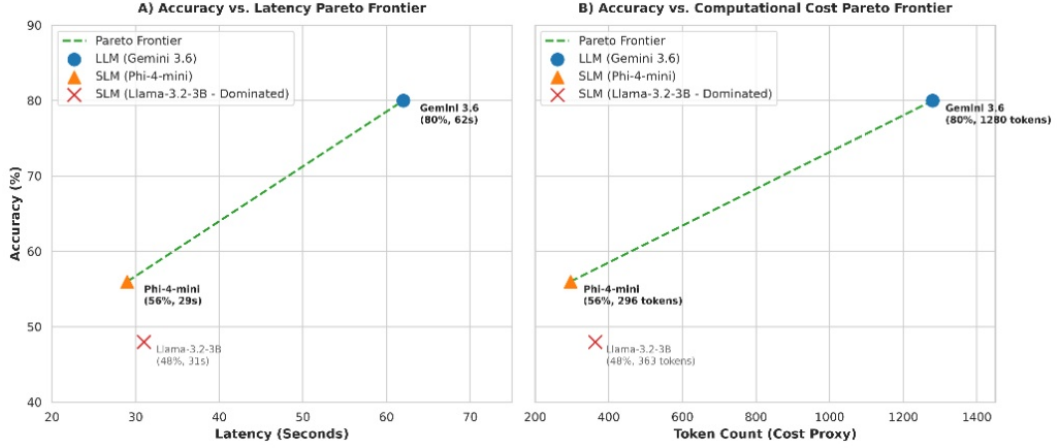
* It is important to note that the AMC12 uses a unique scoring method: out of a total of 150 points, each correct answer nets 6 points, each incorrect answer nets 0 points, and each unanswered question nets 1.5 points. This experiment is focused on ensuring models are evaluated solely on their mathematical reasoning and problem-solving attempts rather than their ability to strategically omit questions for point optimization, so the traditional scoring guideline is not used. All logs and scripts are publicly available for replication [14].

Table 1: AMC12 Benchmark Results Across Evaluated Models

Model	Accuracy	Latency (s)	Token Cost
Gemini 3.6 (LLM)	72%	145	1240
Phi-4-Mini (SLM)	32%	88	622
Llama3.2:3B (SLM)	32%	137	866

Comparison of accuracy, response latency, and token generation cost on AMC12 questions

Figure 1: Model Performance Comparison on AMC12 Benchmark



Accuracy versus latency/token cost trade-offs across Gemini 3.6, Phi-4-Mini, and Llama3.2:3B on AMC12

4 Results

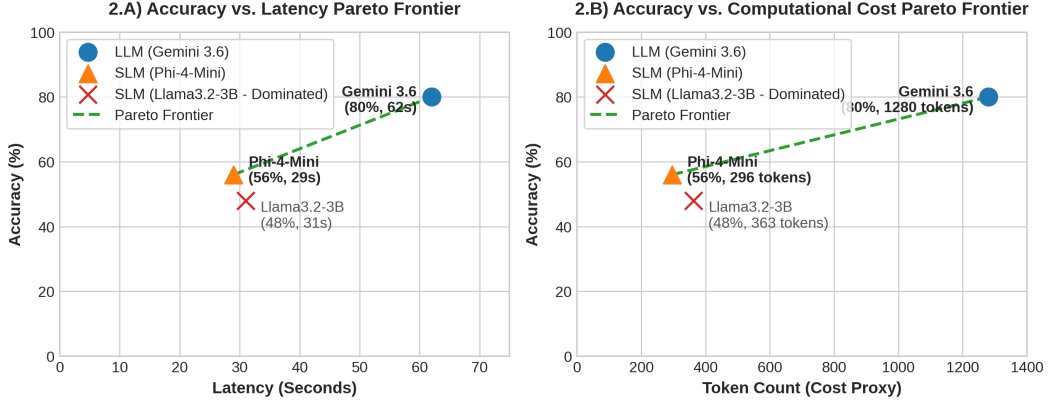
As depicted in Table 1, the LLM generally has a higher accuracy compared to the SLMs by 40%. However, the latency and overall token cost appear to be significantly lower for the SLMs. Specifically, Phi-4-Mini is designed to be an SLM for mathematical reasoning; while it is noticeably less accurate than Gemini 3.6, its lower latency (by 39.3%) and token cost (49.8%) make it a viable option for solving mathematical problems on examinations easier than the AMC12. If sacrificing some accuracy is acceptable, Phi-4-Mini may be an optimal choice to consider.

Table 2: AMC8 Benchmark Results Across Evaluated Models

Model	Accuracy	Latency (s)	Token Cost
Gemini 3.6 (LLM)	80%	62	1280
Phi-4-Mini (SLM)	56%	29	296
Llama3.2:3B (SLM)	48%	31	363

Comparison of accuracy, response latency, and token generation cost on AMC8 questions

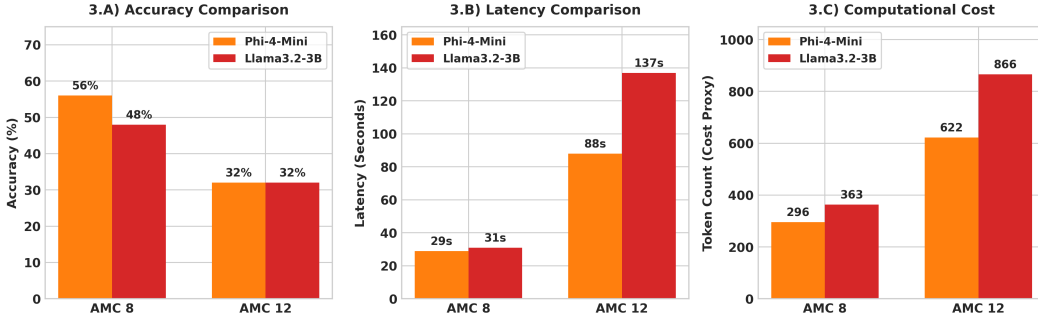
Figure 2: Model Performance Comparison on AMC8 Benchmark



Accuracy versus latency/token cost trade-offs across Gemini 3.6, Phi-4-Mini, and Llama3.2:3B on AMC8

As depicted in Table 2, the LLM achieves higher accuracy than the top-performing SLM by approximately 24%. However, the SLM’s latency and token cost are reduced by 53.2% and 76.9%, respectively, compared to the LLM. This makes it a compelling alternative for mathematical tasks on exams similar in difficulty to the AMC8.

Figure 3: Comparative Analysis: Phi-4-Mini vs. Llama3.2:3B



Direct comparison showing Phi-4-Mini’s Pareto dominance over Llama3.2:3B in accuracy, latency, and token efficiency

5 Discussion and Limitations

This section describes how the experiment’s results can be interpreted along with experimental limitations.

5.1 SLM Practicality

It is imperative to note that, ceteris paribus, higher accuracy, lower latency, and lower computational cost are more favorable and show a model is running more efficiently. In Figure 3, regardless of the exam, Phi-4-Mini had equal to or greater accuracy (56% to 48% on the AMC8 and 32% to 32% on the AMC12), lower latency (31s to 29s on the AMC8 and 88s to 137s on the AMC12), and lower computational cost (296 tokens to 363 tokens on the AMC8 and 622 tokens to 866 tokens on the AMC12) compared to Llama3.2:3B. These results indicate that Phi-4-Mini (the SLM designed for mathematical reasoning tasks) pareto dominates and is more effective than Llama3.2:3B (the SLM designed for textual tasks) at mathematical tasks.

5.2 LLM for Complexity

As shown in figures 1A and 1B, the pareto optimal point is shared between Gemini 3.6 and Phi-4-Mini. Phi-4-Mini occupies the optimal position for resource-constrained environments, achieving 32% accuracy in 88s with 622 tokens, whereas Gemini 3.6 occupies the optimal position when maximal accuracy is required, being 72%. Despite resulting in a higher accuracy, the latency of Gemini 3.6 was significantly penalized by 39.3% (84s vs 145s) and generated 49.8% more tokens (622 vs 1240). This indicates that parameter count alone does not dictate efficiency, and model-specific architectural optimizations play a decisive role. The slope between Phi-4-Mini and Gemini 3.6 shows that going from using an optimal SLM to an LLM yields a 40% accuracy increase, though at the cost of approximately a minute of latency and 600 tokens. For high-complex mathematical examinations such as the AMC12 where accuracy is weighted much heavier than time, this tradeoff favors LLMs, ensuring LLMs still remain dominant over SLMs despite having greater latency and token cost.

5.3 SLM for Simplicity

The marginal slope between Phi-4-Mini and Gemini 3.6 shifts drastically when transitioning from high-complexity to foundational mathematics. Phi-4-Mini once again occupies the optimal position for resource-constrained environments, achieving 56% accuracy in 29s with 296 tokens, whereas Gemini 3.6 occupies the optimal position when maximal accuracy is required, being 80%. It is important to note that in this easier exam, the accuracy difference between Phi-4-Mini and Gemini 3.6 was only 24% (56% vs 80%), though the latency and token count for Phi-4-Mini was significantly more efficient, lower than 53.2% and 76.9% respectively. This notes that for easier mathematical tasks, the accuracy between SLMs tends to be closer towards each other while the latency and computational cost for SLMs is much lower than that of LLMs, hinting at SLMs becoming increasingly more viable.

5.4 Slope for Simplicity

Model performance can be simulated through the linear relationship $y = mx + b$, where y represents accuracy, x represents the resource metric (latency or tokens), b represents baseline accuracy, and the slope m represents the marginal return: $\frac{\Delta \text{accuracy}}{\Delta \text{resource}}$. Through this relationship, compelling quantitative proof is found advocating the use of SLMs for easier mathematical tasks. For the AMC12 benchmark, the slope for token expenditure is approximately $m \approx 0.0647\%$ accuracy per extra token spent ($\frac{1}{m} \approx 15.5$ tokens per 1% accuracy gain). However, for the AMC8 benchmark, marginal returns significantly worsen: the slope for token expenditure is approximately $m \approx 0.0244\%$ accuracy per extra token spent ($\frac{1}{m} \approx 41.0$ tokens per 1% accuracy gain). Noting that Phi-4-Mini was able to replicate 70% of Gemini3.6's success, comparing these benchmarks highlights a noticeable difference in computational return on investment. The overall token expenditure multiplies approximately 2.75x (from 15.5 to 41 tokens per 1% accuracy increase) and the accuracy-to-latency ratio increases approximately 4% on the AMC8 benchmark, greatly showing that on more foundational mathematical tasks such as the AMC8, traditional LLMs operate deep within a region of diminishing returns along with a less efficient accuracy-to-latency ratio.

5.5 Limitations

A few limitations were present in this experiment, one of the most notable ones being that the SLMs tested were run locally. If the local device struggled with CPU/Processing, the SLMs may have had a more difficult time processing information, therefore possibly diminishing accuracy and resulting in higher latency. Additionally, while latency for the SLMs were computed through a simple python script, since Gemini3.6 (LLM) was tested through the Gemini web app, there was no direct indicator of the amount of time the model took to come up and output a response. This latency was hand-timed, and therefore, may be off by a few fractions of a second, though this does not serve to be a huge issue, as latency for the LLM tended to be in the high double-digits/triple digits, where a few fractions of a second could be attributed to various factors such as system/hardware bottlenecks or multinomial sampling. In any case, a few fractions of a second does not add much significance to the overall results.

6 Conclusion

This study investigated the Pareto frontier between accuracy, latency, and token cost across LLMs and SLMs using the AMC12 high school mathematics benchmark. The empirical findings demonstrate a clear capability threshold: while open-weight SLMs like Phi-4-Mini and Llama3.2:3B offer substantial resource savings, reducing generation latency by up to 39.3% and token footprint by nearly 50% compared to frontier models like Gemini 3.6, they suffer a dramatic accuracy penalty on complex reasoning tasks (32% vs. 72%).

These results indicate that for multi-step, competition-grade mathematical reasoning, the accuracy advantage of LLMs justifies their higher computational expense. SLMs in their standalone zero-shot configuration do not yet reach the Pareto optimal boundary for high-complexity problem-solving. Consequently, deploying current SLMs for unassisted, high-stakes quantitative reasoning is not recommended.

However, for more foundational mathematical tasks SLMs are more tailored towards, the accuracy differential tends to not be as large while having significantly lower latency and computational cost. Generally, on foundational tasks, LLMs will be superior in accuracy, but the overall effectiveness when taking into account latency and computational cost justifies the use of SLMs over LLMs.

For future work, researchers should explore hybrid model routing architectures, where lightweight classifiers route low-complexity problems to SLMs while escalating harder queries to cloud-based LLMs. Additionally, investigating whether intermediate efficiency strategies such as Chain-of-Thought prompting or fine-grained test-time compute scaling on SLMs can bridge the 40% accuracy gap without negating their inherent latency benefits remains a promising direction. Furthermore, utilizing more niche benchmarks with a difficulty between the AMC8 and AMC12 may help establish a more precise empirical boundary for optimal model selection.

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